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(54) VEHICLE THERMAL MANAGEMENT SYSTEM

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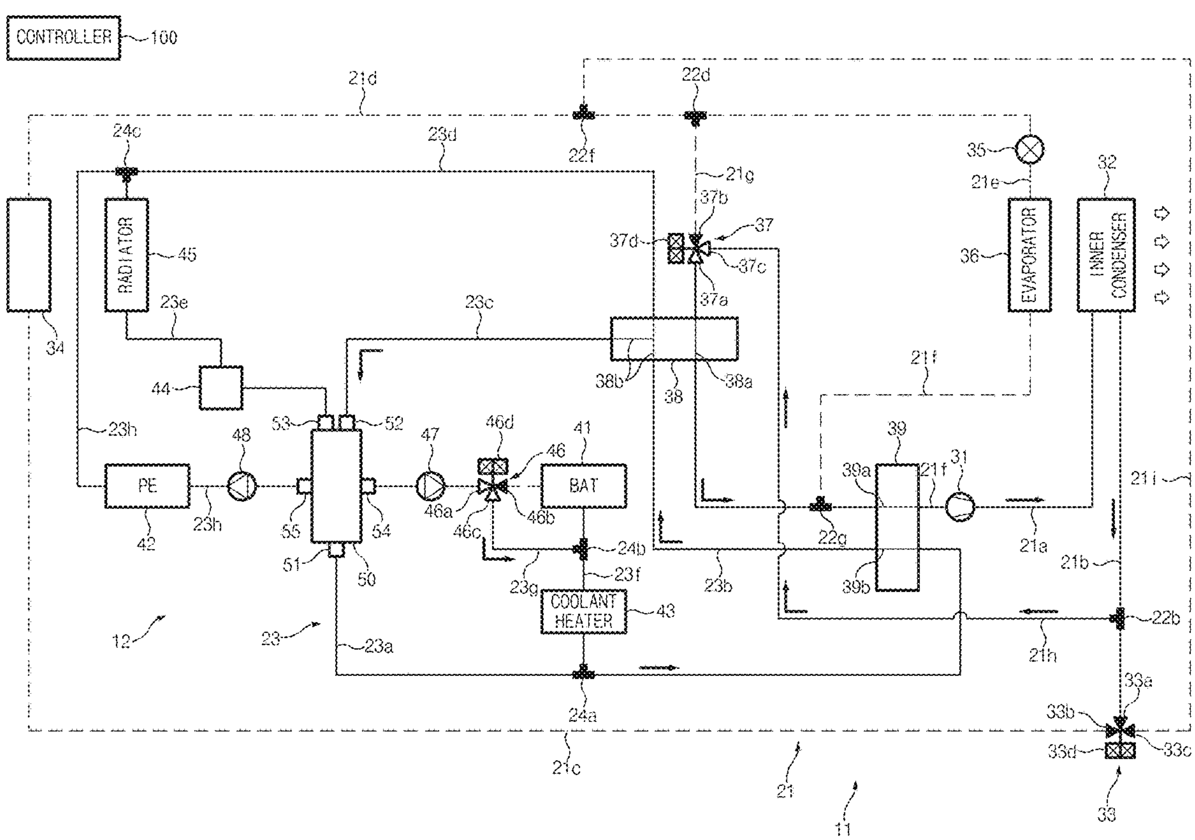
(57) ABSTRACT

A vehicle thermal management system includes a heating, ventilation, and air conditioning (HVAC) subsystem having a compressor, an accumulator disposed on an upstream side of the compressor, and a refrigerant circulation path fluidly connected to the accumulator. The thermal management system also includes a coolant subsystem having a coolant heater and a coolant circulation path fluidly connected to the coolant heater. The accumulator includes a refrigerant passage fluidly connected to the refrigerant circulation path and includes a coolant passage fluidly connected to the coolant circulation path.



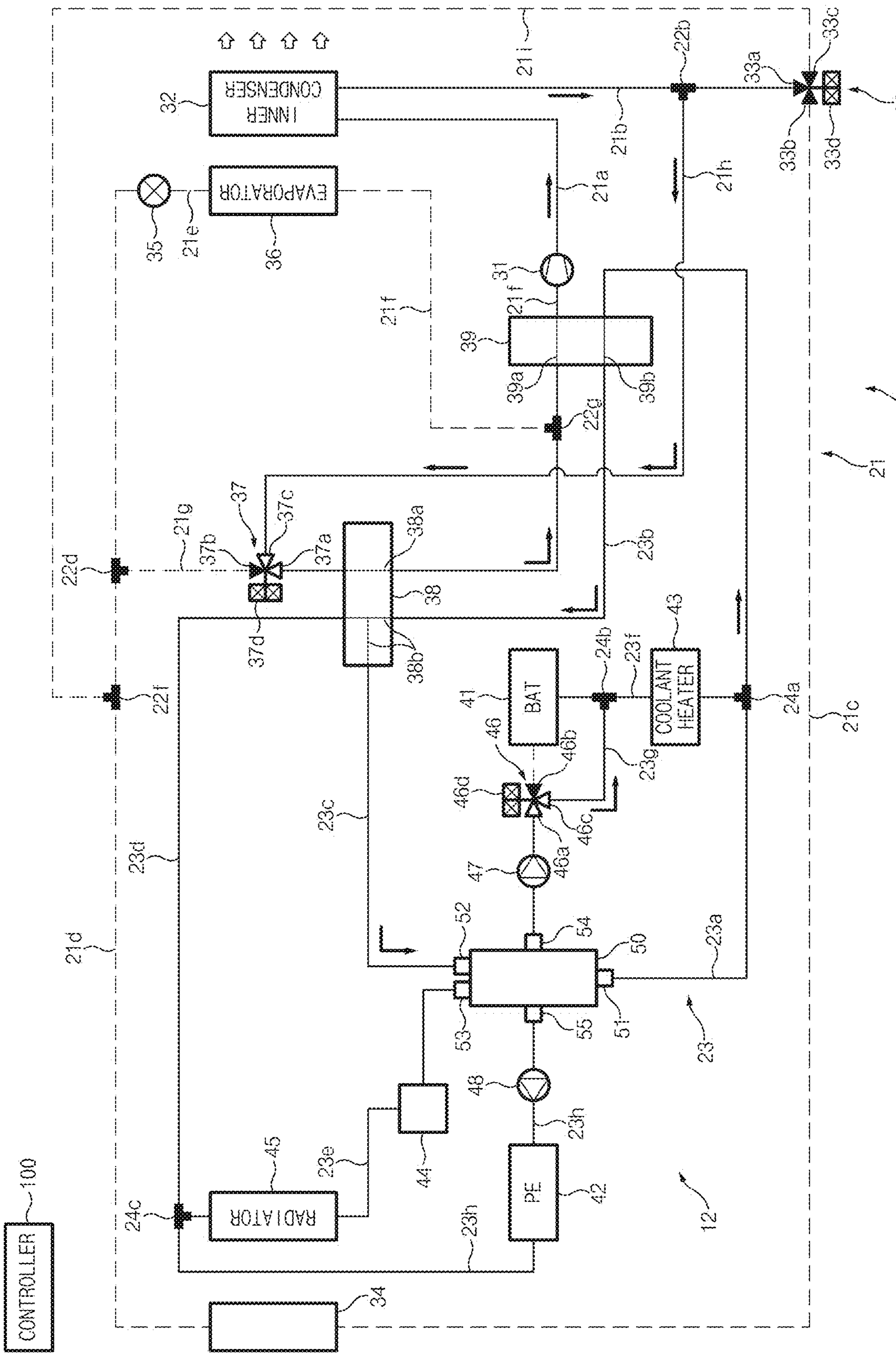

INNER CONDENSER
EVAPORATOR
BAT
COOLANT HEATER
RADIATOR
PE
CONTROLLER
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FIG. 1

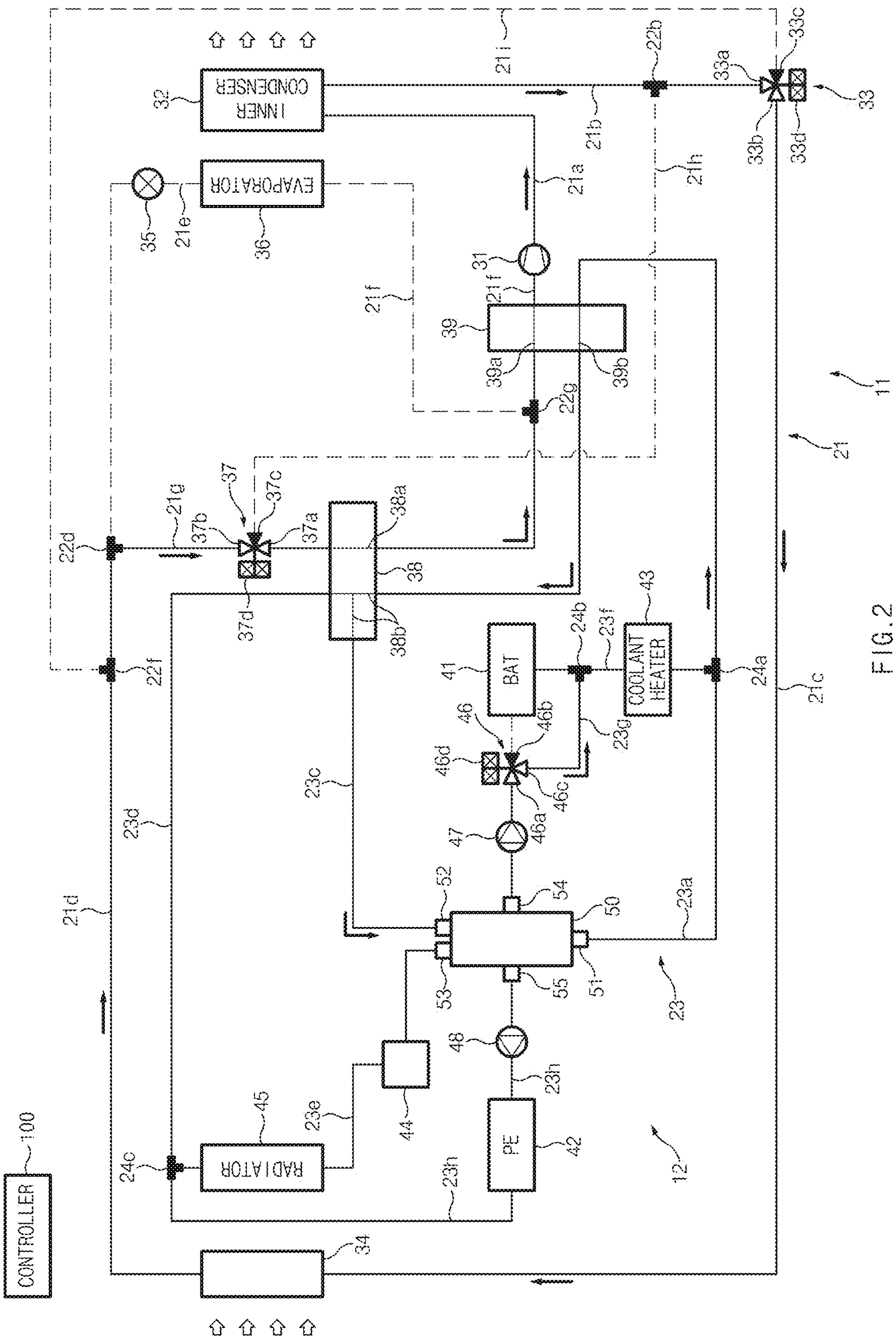
CONTROLLER
RADIATOR
PE
BAT
COOLANT HEATER
EVAPORATOR
INNER CONDENSER

FIG. 2

VEHICLE THERMAL MANAGEMENT SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to Korean Patent Application No. 10-2024-0031502, filed on Mar. 5, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a vehicle thermal management system, and more particularly, to a vehicle thermal management system configured to transfer heat between a coolant heater and an accumulator when an ambient temperature is relatively low.

BACKGROUND

[0003] With a growing interest in energy efficiency and environmental issues, there is a demand for the development of eco-friendly vehicles that can replace internal combustion engine vehicles. Such eco-friendly vehicles are classified into electric vehicles which are driven using fuel cells or electricity as a power source and hybrid vehicles which are driven using an engine and a battery.

[0004] Electric vehicles or hybrid vehicles may include a vehicle thermal management system for heating, ventilation, and air conditioning (HVAC) in a cabin (or passenger compartment) and maintaining a battery and/or power electronics (PE) components of a PE system at appropriate temperatures. The vehicle thermal management system may include an HVAC subsystem for heating, ventilation, and air conditioning the cabin, and a coolant subsystem for maintaining the PE components and/or the battery at appropriate temperatures.

[0005] In the vehicle thermal management system according to the related art, when the HVAC subsystem operates in a heating mode to perform a cabin heating operation in a relatively low ambient temperature condition (e.g., −10° C. or below) such as in winter, an amount of heat absorbed by a refrigerant through an exterior heat exchanger, a battery chiller, and the like may be relatively insufficient. Accordingly, the HVAC subsystem may fail to smoothly operate in the heating mode. Thus, when the ambient temperature is relatively low, an electric heater such as a PTC heater may be used to heat the cabin, resulting in a significant waste of electric energy.

[0006] In addition, the coolant subsystem of the vehicle thermal management system may include a coolant heater, and the coolant heater may be designed to increase a temperature of the battery to a battery operating temperature when the temperature of the battery is lowered below the battery operating temperature. However, the coolant heater may only be used for battery warming-up, so the usability of the coolant heater may be reduced.

[0007] The above information described in this background section is provided to assist in understanding the background of the inventive concept. The above information may include any technical concept that is not considered as the prior art that is already known to those having ordinary skill in the art.

SUMMARY

[0008] The present disclosure has been made to solve the above-mentioned problems occurring in the prior art while advantages achieved by the prior art are maintained intact.

[0009] An aspect of the present disclosure provides a vehicle thermal management system configured to transfer heat between a coolant heater and an accumulator when an ambient temperature is relatively low, thereby improving heating performance.

[0010] According to an aspect of the present disclosure, a vehicle thermal management system may include a heating, ventilation, and air conditioning (HVAC) subsystem having a compressor, an accumulator disposed on an upstream side of the compressor, and a refrigerant circulation path fluidly connected to the accumulator. The system may also include a coolant subsystem having a coolant heater and a coolant circulation path fluidly connected to the coolant heater. The accumulator may include a refrigerant passage fluidly connected to the refrigerant circulation path, and a coolant passage fluidly connected to the coolant circulation path.

[0011] The vehicle thermal management system may further include a battery chiller having a refrigerant passage fluidly connected to the refrigerant circulation path, and a coolant passage fluidly connected to the coolant circulation path.

[0012] The refrigerant passage of the battery chiller may be located on an upstream side of the refrigerant passage of the accumulator, and the coolant passage of the battery chiller may be located on a downstream side of the coolant passage of the accumulator.

[0013] The coolant heater may be located on an upstream side of the coolant passage of the accumulator.

[0014] The HVAC subsystem may further include an interior condenser disposed on a downstream side of the compressor, a refrigerant control valve disposed on a downstream side of the interior condenser, an exterior heat exchanger disposed on a downstream side of the refrigerant control valve, and a chiller-side expansion valve disposed on the upstream side of the refrigerant passage of the battery chiller. The chiller-side expansion valve may be configured to allow a refrigerant to flow from any one of the exterior heat exchanger and the interior condenser to the refrigerant passage of the battery chiller.

[0015] The refrigerant circulation path may include a heating-side bypass line extending from an upstream point of the refrigerant control valve to an upstream point of the refrigerant passage of the battery chiller.

[0016] The chiller-side expansion valve may include a first port fluidly communicating with the refrigerant passage of the battery chiller, a second port fluidly communicating with the exterior heat exchanger, and a third port fluidly communicating with the heating-side bypass line.

[0017] The chiller-side expansion valve may be configured to allow the first port to be fluidly connected to at least one of the second port and the third port by an actuator.

[0018] The coolant subsystem may further include a battery and a coolant control valve fluidly connected to the coolant circulation path. The coolant control valve may be configured to control the flow of a coolant passing through the battery, the coolant heater, the coolant passage of the accumulator, and the coolant passage of the battery chiller.

[0019] The coolant heater may be located on a downstream side of the battery.

[0020] The coolant subsystem may further include a battery bypass line extending from an upstream point of the battery to a downstream point of the battery.

[0021] The coolant subsystem may further include a distribution valve configured to allow the coolant discharged from the coolant control valve to be directed to at least one of the battery and the battery bypass line.

[0022] The distribution valve may include an inlet port fluidly communicating with the coolant control valve, a first outlet port fluidly communicating with the battery, and a second outlet port fluidly communicating with the battery bypass line.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The above and other objects, features, and advantages of the present disclosure should be more apparent from the following detailed description taken in conjunction with the accompanying drawings:

[0024] FIG. **1** illustrates a process of a vehicle thermal management system according to an embodiment of the present disclosure in which a refrigerant absorbs heat through an accumulator and a battery chiller; and

[0025] FIG. **2** illustrates a process of a vehicle thermal management system according to an embodiment of the present disclosure in which a refrigerant absorbs heat through an accumulator, a battery chiller, and an exterior heat exchanger.

DETAILED DESCRIPTION

[0026] Hereinafter, embodiments of the present disclosure are described in detail with reference to the accompanying drawings. In the drawings, the same reference numerals are used throughout to designate the same or equivalent elements. In addition, a detailed description of well-known techniques associated with the present disclosure has been omitted in order not to unnecessarily obscure the gist of the present disclosure.

[0027] Terms such as first, second, A, B, (a), and (b) may be used to describe the elements in the embodiments of the present disclosure. These terms are only used to distinguish one element from another element, and the intrinsic features, sequence or order, and the like of the corresponding elements are not limited by the terms. Unless otherwise defined, all terms used herein, including technical or scientific terms, have the same meanings as those generally understood by those having ordinary skill in the art to which the present disclosure belongs. Such terms as those defined in a generally used dictionary are to be interpreted as having meanings consistent with the contextual meanings in the relevant field of art. Such terms are not to be interpreted as having ideal or excessively formal meanings unless clearly defined as having such in the present application.

[0028] When a controller, component, device, element, part, unit, module, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the controller, component, device, element, part, unit, or module should be considered herein as being "configured to" meet that purpose or perform that operation or function. Each controller, component, device, element, part, unit, module, and the like may separately embody or be included with a processor and a memory, such as a non-transitory computer-readable media, as part of the apparatus.

[0029] Referring to FIGS. **1** and **2**, a vehicle thermal management system according to an embodiment of the present disclosure may include a heating, ventilation, and air conditioning (HVAC) subsystem **11** configured to heat or cool air in a cabin of a vehicle. The vehicle thermal management system may also include a coolant subsystem **12** configured to maintain a battery **41** and/or a power electronics (PE) component **42** at appropriate temperatures.

[0030] The HVAC subsystem **11** may include a refrigerant circulation path **21** through which a refrigerant circulates. The refrigerant circulation path **21** may be fluidly connected to a compressor **31**, an interior condenser **32**, a refrigerant control valve **33**, an exterior heat exchanger **34**, a cooling-side expansion valve **35**, an evaporator **36**, a chiller-side expansion valve **37**, a battery chiller **38**, and an accumulator **39**.

[0031] Referring to FIG. **1**, the refrigerant circulation path **21** may include: a first refrigerant line **21*a*** extending from an outlet of the compressor **31** to the interior condenser **32**; a second refrigerant line **21*b*** extending from the interior condenser **32** to the refrigerant control valve **33**; a third refrigerant line **21*c*** extending from the refrigerant control valve **33** to an inlet of the exterior heat exchanger **34**; a fourth refrigerant line **21*d*** extending from an outlet of the exterior heat exchanger **34** to an inlet of the cooling-side expansion valve **35**; a fifth refrigerant line **21*e*** extending from an outlet of the cooling-side expansion valve **35** to an inlet of the evaporator **36**; and a sixth refrigerant line **21*f*** extending from an outlet of the evaporator **36** to an inlet of the compressor **31**.

[0032] The refrigerant circulation path **21** may also include a distribution line **21*g*** extending from a branch point **22*d*** of the fourth refrigerant line **21*d*** to a joint point **22*g*** of the sixth refrigerant line **21*f***. An inlet of the distribution line **21*g*** may be fluidly connected to the branch point **22*d*** of the fourth refrigerant line **21*d*** on the upstream side of the cooling-side expansion valve **35**. An outlet of the distribution line **21*g*** may be fluidly connected to the joint point **22*g*** of the sixth refrigerant line **21*f*** on the upstream side of the compressor **31**.

[0033] The refrigerant circulation path **21** may include a heating-side bypass line **21*h*** extending from a branch point **22*b*** of the second refrigerant line **21*b*** to one point of the distribution line **21*g***. The heating-side bypass line **21*h*** may extend from an upstream point of the refrigerant control valve **33** to an upstream point of a refrigerant passage **38*a*** of the battery chiller **38**. An inlet of the heating-side bypass line **21*h*** may be fluidly connected to the branch point **22*b*** of the second refrigerant line **21*b*** on the upstream side of the refrigerant control valve **33**, and an outlet of the heating-side bypass line **21*h*** may be fluidly connected to a third port **37*c*** of the chiller-side expansion valve **37** on the upstream side of the refrigerant passage **38*a*** of the battery chiller **38**.

[0034] The refrigerant circulation path **21** may include a dehumidifying-side bypass line **21*i*** extending from the refrigerant control valve **33** to a joint point **22*f*** of the fourth refrigerant line **21*d***. An inlet of the dehumidifying-side bypass line **21*i*** may be fluidly connected to a third port **33*c*** of the refrigerant control valve **33**. Furthermore, an outlet of the dehumidifying-side bypass line **21*i*** may be fluidly connected to the joint point **22*f*** of the fourth refrigerant line **21*d*** on the upstream side of the cooling-side expansion valve **35**.

[0035] The refrigerant circulation path **21** may allow the flow of the refrigerant to vary depending on an operating mode (a heating mode, a cooling mode, a dehumidifying mode, and the like) of the HVAC subsystem **11**, a cooling mode and a warming-up mode of the battery **41**, a cooling mode of the PE component **42**, and the like.

[0036] The compressor **31** may compress the refrigerant. According to an embodiment, the compressor **31** may be an electric compressor driven by electric energy.

[0037] The interior condenser **32** may be configured to condense the refrigerant received from the compressor **31**. In other words, the refrigerant compressed by the compressor **31** may transfer heat to the air and be condensed in the interior condenser **32**. Accordingly, the interior condenser **32** may heat the air using the refrigerant compressed by the compressor **31**, and the air heated by the interior condenser **32** may be directed into the cabin.

[0038] The refrigerant control valve **33** may be configured to control the flow of the refrigerant in a manner that allows the refrigerant discharged from the interior condenser **32** to be directed to at least one of the exterior heat exchanger **34** and the dehumidifying-side bypass line **21***i*. The refrigerant control valve **33** may include a first port **33***a* fluidly communicating with the interior condenser **32**, a second port **33***b* fluidly communicating with the exterior heat exchanger **34**, and the third port **33***c* fluidly communicating with the battery chiller **38** and the evaporator **36**. The refrigerant control valve **33** may be configured to allow the first port **33***a* to communicate with at least one of the second port **33***b* and the third port **33***c* by an actuator **33***d* or to close all of the first port **33***a*, the second port **33***b*, and the third port **33***c*. For example, the refrigerant control valve **33** may be designed to move a valve member such as a ball member or a needle member in a valve housing to thereby allow the first port **33***a* to be fluidly connected to at least one of the second port **33***b* and the third port **33***c* or to close all of the first port **33***a*, the second port **33***b*, and the third port **33***c*. The valve member may be moved by the actuator **33***d*.

[0039] When the refrigerant control valve **33** performs a first switching operation to allow the first port **33***a* to be fluidly connected to the second port **33***b*, the refrigerant discharged from the interior condenser **32** may be directed to the exterior heat exchanger **34** through the first port **33***a* and the second port **33***b*. When the refrigerant control valve **33** performs the first switching operation, the opening degree of the second port **33***b* may be adjusted by the valve member so that the refrigerant may or may not be expanded. According to an embodiment, when the refrigerant control valve **33** performs the first switching operation, the opening degree of the second port **33***b* may be adjusted by the valve member to exceed 0% and be less than 100% so that the refrigerant discharged from the second port **33***b* of the refrigerant control valve **33** may be expanded. According to another embodiment, when the refrigerant control valve **33** performs the first switching operation, the opening degree of the second port **33***b* may be adjusted to 100% by the valve member (i.e., the second port **33***b* may be fully opened) so that the refrigerant discharged from the second port **33***b* of the refrigerant control valve **33** may not be expanded.

[0040] When the refrigerant control valve **33** performs a second switching operation to allow the first port **33***a* to be fluidly connected to the third port **33***c*, the refrigerant discharged from the interior condenser **32** may be directed to the refrigerant passage **38***a* of the battery chiller **38** and/or the evaporator **36** through the first port **33***a* and the third port **33***c*, thereby bypassing the exterior heat exchanger **34**. When the refrigerant control valve **33** performs the second switching operation, the opening degree of the third port **33***c* may be adjusted to 100% by the valve member (i.e., the third port **33***c* may be fully opened) so that the refrigerant discharged from the third port **33***c* of the refrigerant control valve **33** may not be expanded.

[0041] When the refrigerant control valve **33** performs a third switching operation to close the first port **33***a*, the second port **33***b*, and the third port **33***c*, the refrigerant discharged from the interior condenser **32** may be directed to the third port **37***c* of the chiller-side expansion valve **37** through the heating-side bypass line **21***h*.

[0042] The exterior heat exchanger **34** may be disposed adjacent to a front grille of the vehicle, and the exterior heat exchanger **34** may be exposed to the ambient air so that heat may be transferred between the exterior heat exchanger **34** and the ambient air. The exterior heat exchanger **34** may be configured to condense the refrigerant received from the interior condenser **32**. An active air flap (not shown) may be provided to open or close the front grille of the vehicle. In particular, the exterior heat exchanger **34** may exchange heat with the ambient air forcibly blown by a cooling fan (not shown) so that a heat transfer rate between the exterior heat exchanger **34** and the ambient air may be further increased. During a cooling operation of the HVAC subsystem **11**, the exterior heat exchanger **34** may be configured to condense the refrigerant received from the interior condenser **32**. In other words, the exterior heat exchanger **34** may serve as a condenser that condenses the refrigerant during the cooling operation of the HVAC subsystem **11**. During a heating operation of the HVAC subsystem **11**, the exterior heat exchanger **34** may serve as an evaporator that evaporates the refrigerant.

[0043] The cooling-side expansion valve **35** may be disposed between the exterior heat exchanger **34** and the evaporator **36** in the refrigerant circulation path **21**. The cooling-side expansion valve **35** may be disposed on the upstream side of the evaporator **36** so that it may adjust the flow of the refrigerant and/or the flow rate of the refrigerant into the evaporator **36**, and the cooling-side expansion valve **35** may be configured to expand the refrigerant received from the exterior heat exchanger **34**. The cooling-side expansion valve **35** may be a thermal expansion valve (TXV) which senses the temperature and/or pressure of the refrigerant and adjusts the opening degree of the cooling-side expansion valve **35**.

[0044] According to an embodiment of the present disclosure, the cooling-side expansion valve **35** may be a TXV having a shut-off valve selectively blocking or unblocking the flow of the refrigerant into an internal passage of the cooling-side expansion valve **35**, and the shut-off valve may be a solenoid valve. As a controller **100** controls the shut-off valve, the shut-off valve may be opened or closed so that the shut-off valve may unblock or block the flow of the refrigerant into the cooling-side expansion valve **35**. When the shut-off valve is opened, the refrigerant may be allowed to flow into the cooling-side expansion valve **35**. Additionally, when the shut-off valve is closed, the refrigerant may be blocked from flowing into the cooling-side expansion valve **35**. According to an embodiment, the shut-off valve may be mounted in a valve body of the cooling-side expansion valve **35**, thereby opening or closing the internal passage of the cooling-side expansion valve **35**. According to another embodiment, the shut-off valve may be disposed on the upstream side of the cooling-side expansion valve **35**, thereby selectively opening or closing an inlet of the cooling-side expansion valve **35**.

[0045] When the shut-off valve is closed, the cooling-side expansion valve **35** may be blocked. Accordingly, the refrigerant may not be directed to the cooling-side expansion valve **35** and the evaporator **36**, but may only be directed to the battery chiller **38**. In other words, when the shut-off valve is closed, the cooling operation of the HVAC subsystem **11** may not be performed. When the shut-off valve is opened, the refrigerant may be directed to the cooling-side expansion valve **35** and the evaporator **36**. In other words, when the shut-off valve is opened, the cooling operation of the HVAC subsystem **11** may be performed.

[0046] The evaporator **36** may be configured to cool the air using the refrigerant expanded by the cooling-side expansion valve **35**.

[0047] The battery chiller **38** may be fluidly connected to the distribution line **21***g*. The battery chiller **38** may be configured to transfer heat between the refrigerant passing through the distribution line **21***g* and a coolant passing through a coolant circulation path **23**. The battery chiller **38** may include the refrigerant passage **38***a* through which the refrigerant passes, and a coolant passage **38***b* through which the coolant passes. The refrigerant passage **38***a* may be fluidly connected to the distribution line **21***g*, and the coolant passage **38***b* may be fluidly connected to the coolant circulation path **23**. The refrigerant passage **38***a* and the coolant passage **38***b* may be adjacent to each other or contact each other in the battery chiller **38**, and the refrigerant passage **38***a* may be fluidly separated from the coolant passage **38***b*. Accordingly, the battery chiller **38** may be configured to transfer heat between the coolant passing through the coolant passage **38***b* and the refrigerant passing through the refrigerant passage **38***a*. The refrigerant passage **38***a* of the battery chiller **38** may be located on the upstream side of the compressor **31**. Accordingly, the refrigerant discharged from the refrigerant passage **38***a* of the battery chiller **38** may be directed to the compressor **31**.

[0048] The chiller-side expansion valve **37** may be disposed on the upstream side of the refrigerant passage **38***a* of the battery chiller **38** on the distribution line **21***g*. The chiller-side expansion valve **37** may be configured to control the flow of the refrigerant in a manner that allows the refrigerant to flow from any one of the exterior heat exchanger **34** and the interior condenser **32** to the refrigerant passage **38***a* of the battery chiller **38**. In particular, the chiller-side expansion valve **37** may adjust the flow of the refrigerant and/or the flow rate of the refrigerant into the refrigerant passage **38***a* of the battery chiller **38**. Additionally, the chiller-side expansion valve **37** may expand the refrigerant received from the exterior heat exchanger **34** or the interior condenser **32**.

[0049] The chiller-side expansion valve **37** may include a first port **37***a* fluidly communicating with the refrigerant passage **38***a* of the battery chiller **38**, a second port **37***b* fluidly communicating with the exterior heat exchanger **34**, and the third port **37***c* fluidly communicating with the heating-side bypass line **21***h*.

[0050] The chiller-side expansion valve **37** may be configured to allow the first port **37***a* to communicate with at least one of the second port **37***b* and the third port **37***c* by an actuator **37***d*. For example, the chiller-side expansion valve **37** may be designed to move a valve member such as a ball member or a needle member in a valve housing to thereby allow the first port **37***a* to be fluidly connected to at least one of the second port **37***b* and the third port **37***c*. The valve member may be moved by the actuator **37***d*.

[0051] When the chiller-side expansion valve **37** performs a first switching operation to allow the first port **37***a* to be fluidly connected to the second port **37***b*, the refrigerant discharged from the exterior heat exchanger **34** may be directed to the refrigerant passage **38***a* of the battery chiller **38** through the second port **37***b* and the first port **37***a*. When the chiller-side expansion valve **37** performs the first switching operation, the opening degree of the first port **37***a* may be adjusted by the valve member so that the refrigerant may be expanded. When the chiller-side expansion valve **37** performs the first switching operation, the opening degree of the first port **37***a* may be adjusted by the valve member to exceed 0% and be less than 100% so that the refrigerant discharged from the first port **37***a* of the chiller-side expansion valve **37** may be expanded.

[0052] When the chiller-side expansion valve **37** performs a second switching operation to allow the first port **37***a* to be fluidly connected to the third port **37***c*, the refrigerant discharged from the interior condenser **32** may be directed to the refrigerant passage **38***a* of the battery chiller **38** through the heating-side bypass line **21***h* and the third port **37***c* and the first port **37***a* of the chiller-side expansion valve **37**. As a result, the refrigerant bypasses the exterior heat exchanger **34**. When the chiller-side expansion valve **37** performs the second switching operation, the opening degree of the first port **37***a* may be adjusted by the valve member to exceed 0% and be less than 100% so that the refrigerant discharged from the first port **37***a* of the chiller-side expansion valve **37** may be expanded.

[0053] The HVAC subsystem **11** may further include the accumulator **39** disposed on the upstream side of the compressor **31** in the refrigerant circulation path **21**. The accumulator **39** may be configured to separate a liquid refrigerant from the refrigerant received from the refrigerant passage **38***a* of the battery chiller **38** and/or the evaporator **36**, thereby preventing the liquid refrigerant from flowing into the compressor **31**.

[0054] According to an embodiment of the present disclosure, the accumulator **39** may include a refrigerant passage **39***a* through which the refrigerant passes, and a coolant passage **39***b* through which the coolant passes. The refrigerant passage **39***a* may be fluidly connected to the sixth refrigerant line **21***f* of the refrigerant circulation path **21**, and the coolant passage **39***b* may be fluidly connected to the coolant circulation path **23** of the coolant subsystem **12**. The accumulator **39** may serve as a heat exchanger that transfers heat between the coolant passing through the coolant passage **39***b* and the refrigerant passing through the refrigerant passage **39***a*.

[0055] The refrigerant passage **38***a* of the battery chiller **38** may be located on the upstream side of the refrigerant passage **39***a* of the accumulator **39**. Accordingly, the refrigerant discharged from the refrigerant passage **38***a* of the battery chiller **38** may be directed to the refrigerant passage **39***a* of the accumulator **39**. The coolant passage **38***b* of the battery chiller **38** may be located on the downstream side of the coolant passage **39***b* of the accumulator **39**. Accordingly, the coolant discharged from the coolant passage **39***b* of the accumulator **39** may be directed to the coolant passage **38***b* of the battery chiller **38**.

[0056] As described above, the refrigerant circulation path **21** of the HVAC subsystem **11** may be fluidly connected to the refrigerant passage **38***a* of the battery chiller **38** and the refrigerant passage **39***a* of the accumulator **39** so that the refrigerant may sequentially pass through the refrigerant passage **38***a* of the battery chiller **38** and the refrigerant passage **39***a* of the accumulator **39** and then be directed to the compressor **31**.

[0057] The coolant subsystem **12** may include the coolant circulation path **23** through which the coolant circulates. The coolant circulation path **23** may be fluidly connected to the battery **41**, the PE component **42**, a coolant heater **43**, a reservoir **44**, a coolant control valve **50**, a radiator **45**, the coolant passage **39***b* of the accumulator **39**, and the coolant passage **38***b* of the battery chiller **38**.

[0058] The coolant circulation path **23** may allow the flow of the coolant to vary depending on an operating mode (a heating mode, a cooling mode, a dehumidifying mode, and the like) of the HVAC subsystem **11**, a cooling mode and a warming-up mode of the battery **41**, a cooling mode of the PE component **42**, and the like.

[0059] According to an embodiment, the coolant circulation path **23** may include a first coolant line **23***a* extending from a first port **51** of the coolant control valve **50** to the coolant passage **39***b* of the accumulator **39**, a second coolant line **23***b* extending from the coolant passage **39***b* of the accumulator **39** to the coolant passage **38***b* of the battery chiller **38**, and a third coolant line **23***c* extending from the coolant passage **38***b* of the battery chiller **38** to a second port **52** of the coolant control valve **50**. The coolant circulation path **23** may also include a fourth coolant line **23***d* extending from the coolant passage **38***b* of the battery chiller **38** to the radiator **45**, a fifth coolant line **23***e* extending from the radiator **45** to a third port **53** of the coolant control valve **50**, and a sixth coolant line **23***f* extending from a fourth port **54** of the coolant control valve **50** to a joint point **24***a* of the first coolant line **23***a*. Additionally, the coolant circulation path **23** may also include a seventh coolant line **23***h* extending from a fifth port **55** of the coolant control valve **50** to a joint point **24***c* of the fourth coolant line **23***d*, and a battery bypass line **23***g* extending from an upstream point of the battery **41** to a downstream point **24***b* of the battery **41** on the sixth coolant line **23***f*.

[0060] The battery **41** may have a coolant passage provided inside or outside thereof. As the coolant passes through the coolant passage of the battery **41**, the coolant may be heated or cooled, and the battery **41** may be maintained at an appropriate temperature. For example, the battery **41** may be a high-voltage battery pack of an electric vehicle. The battery **41** may be disposed on the sixth coolant line **23***f*.

[0061] The PE component **42** may have a coolant passage provided inside or outside thereof. As the coolant passes through the coolant passage of the PE component **42**, the coolant may be heated or cooled, and the PE component **42** may be maintained at an appropriate temperature. For example, the PE component **42** may be an electric motor which is a driving source of an electric vehicle, an inverter, and the like. The PE component **42** may be disposed on the seventh coolant line **23***h*.

[0062] The coolant heater **43** may have a coolant passage provided inside or outside thereof, and the coolant heater **43** may be an electric heater that heats the coolant. The coolant heater **43** may be disposed on the sixth coolant line **23***f*, and the coolant heater **43** may be disposed on the downstream side of the battery **41**. The coolant heater **43** may be located on the upstream side of the coolant passage **39***b* of the accumulator **39**, and the coolant heated by the coolant heater **43** may sequentially pass through the coolant passage **39***b* of the accumulator **39** and the coolant passage **38***b* of the battery chiller **38**.

[0063] The radiator **45** may be disposed adjacent to the front grille of the vehicle, and the radiator **45** may have a coolant passage provided therein. The coolant passing through the coolant passage of the radiator **45** may be cooled by the ambient air passing by an exterior surface of the radiator **45**. The radiator may cool the coolant using the ambient air forcibly blown by a cooling fan (not shown). The reservoir **44** and the radiator **45** may be disposed on the fifth coolant line **23***e*, and the reservoir **44** may be disposed on the downstream side of the radiator **45**.

[0064] The coolant control valve **50** may be configured to control the flow of the coolant passing through the coolant passage **39***b* of the accumulator **39**, the coolant passage **38***b* of the battery chiller **38**, the battery **41**, the PE component **42**, the coolant heater **43**, and the radiator **45**. According to an embodiment, the coolant control valve **50** may include the first port **51** fluidly communicating with the coolant passage **39***b* of the accumulator **39**, the second port **52** fluidly communicating with the coolant passage **38***b* of the battery chiller **38**, the third port **53** fluidly communicating with the radiator **45**, the fourth port **54** fluidly communicating with the battery **41**, and the fifth port **55** fluidly communicating with the PE component **42**. The coolant control valve **50** may be configured to operate in various switching modes to allow the first port **51**, the second port **52**, the third port **53**, the fourth port **54**, and the fifth port **55** to be fluidly connected to each other.

[0065] The coolant subsystem **12** may include a first pump **47** fluidly communicating with the fourth port **54** of the coolant control valve **50**. The first pump **47** may be disposed on the sixth coolant line **23***f*, and the first pump **47** may be disposed on the upstream side of the battery **41**.

[0066] The coolant subsystem **12** may include a second pump **48** fluidly communicating with the fifth port **55** of the coolant control valve **50**. The second pump **48** may be disposed on the seventh coolant line **23***h*, and the second pump **48** may be disposed on the upstream side of the PE component **42**.

[0067] The coolant subsystem **12** may include a distribution valve **46** configured to control the flow of the coolant in a manner that allows the coolant discharged from the fourth port **54** of the coolant control valve **50** to be directed to at least one of the battery **41** and the battery bypass line **23***g*. The distribution valve **46** may be disposed at a connection point of the battery bypass line **23***g* and the sixth coolant line **23***f*. In other words, the distribution valve **46** may be located at a point at which the coolant is distributed to the battery **41** and the battery bypass line **23***g*. According to an embodiment, the distribution valve **46** may include an inlet port **46***a* fluidly communicating with the fourth port **54** of the coolant control valve **50**, a first outlet port **46***b* fluidly communicating with the battery **41**, and a second outlet port **46***c* fluidly communicating with the battery bypass line **23***g*.

[0068] The distribution valve **46** may be configured to allow the inlet port **46***a* to be fluidly connected to at least one of the first outlet port **46***b* and the second outlet port **46***c* by an actuator **46***d*. For example, the distribution valve **46** may be designed to move a valve member such as a ball member or a needle member in a valve housing to thereby allow the inlet port **46***a* to communicate with at least one of the first outlet port **46***b* and the second outlet port **46***c*. The valve member may be moved by the actuator **46***d*.

[0069] When the distribution valve **46** performs a first switching operation to allow the inlet port **46***a* to be fluidly connected to the first outlet port **46***b*, the coolant discharged from the fourth port **54** of the coolant control valve **50** may be directed to the coolant passage of the battery **41** through the inlet port **46***a* and the first outlet port **46***b*.

[0070] When the distribution valve **46** performs a second switching operation to allow the inlet port **46***a* to be fluidly connected to the second outlet port **46***c*, the coolant discharged from the fourth port **54** of the coolant control valve **50** may be directed to the battery bypass line **23***g* through the inlet port **46***a* and the second outlet port **46***c*.

[0071] The controller **100** may be configured to control the operations of the compressor **31**, the actuator **33***d* of the refrigerant control valve **33**, the actuator **37***d* of the chiller-side expansion valve **37**, the coolant control valve **50**, the first pump **47**, the second pump **48**, the actuator **46***d* of the distribution valve **46**, and the like depending on an operating mode (a heating mode, a cooling mode, a dehumidifying mode, and the like) of the HVAC subsystem **11**, a cooling mode and a warming-up mode of the battery **41**, a cooling mode of the PE component **42**, and the like.

[0072] As described above, the coolant circulation path **23** may be fluidly connected to the coolant heater **43**, the coolant passage **39***b* of the accumulator **39**, and the coolant passage **38***b* of the battery chiller **38**. As a result, the coolant heated by the coolant heater **43** may sequentially pass through the coolant passage **39***b* of the accumulator **39** and the coolant passage **38***b* of the battery chiller **38**. Accordingly, the heated coolant may release heat to the refrigerant passing through the refrigerant passage **39***a* of the accumulator **39** and the refrigerant passing through the refrigerant passage **38***a* of the battery chiller **38**. Thus, the refrigerant may absorb heat in two steps by the battery chiller **38** and the accumulator **39** so that heat absorption and evaporation of the refrigerant may be significantly improved.

[0073] FIG. **1** illustrates a process of a vehicle thermal management system according to an embodiment of the present disclosure in which a refrigerant absorbs heat through an accumulator and a battery chiller.

[0074] Referring to FIG. **1**, in a condition in which an ambient temperature is relatively low (i.e., in winter, in a cold area, and the like), the HVAC subsystem **11** may operate in a heating mode. The refrigerant control valve **33** may perform the third switching operation to close all of the first port **33***a*, the second port **33***b*, and the third port **33***c*. The chiller-side expansion valve **37** may perform the second switching operation to allow the first port **37***a* to be fluidly connected to the third port **37***c*. The cooling-side expansion valve **35** may be blocked. The refrigerant compressed by the compressor **31** may pass through the interior condenser **32** and the refrigerant discharged from the interior condenser **32** may pass through the third port **37***c* and the first port **37***a* of the chiller-side expansion valve **37** through the heating-side bypass line **21***h*. The refrigerant passing through the first port **37***a* of the chiller-side expansion valve **37** may be expanded and the expanded refrigerant may pass through the refrigerant passage **38***a* of the battery chiller **38**. The refrigerant discharged from the refrigerant passage **38***a* of the battery chiller **38** may pass through the refrigerant passage **39***a* of the accumulator **39** and the refrigerant discharged from the refrigerant passage **39***a* of the accumulator **39** may flow into the compressor **31**.

[0075] Referring to FIG. **1**, the coolant control valve **50** may operate to allow the second port **52** to be fluidly connected to the fourth port **54**. The distribution valve **46** may perform the second switching operation to allow the inlet port **46***a* to be fluidly connected to the second outlet port **46***c*. The coolant may be directed to the coolant heater **43** through the inlet port **46***a*, the second outlet port **46***c* of the distribution valve **46**, and the battery bypass line **23***g* by the first pump **47**. As the coolant heater **43** operates, the coolant may be heated by the coolant heater **43**, and the heated coolant may pass through the coolant passage **39***b* of the accumulator **39**. The coolant discharged from the coolant passage **39***b* of the accumulator **39** may pass through the coolant passage **38***b* of the battery chiller **38**. The coolant discharged from the coolant passage **38***b* of the battery chiller **38** may be directed to the second port **52** of the coolant control valve **50**.

[0076] As described above, the coolant heated by the coolant heater **43** may sequentially pass through the coolant passage **39***b* of the accumulator **39** and the coolant passage **38***b* of the battery chiller **38**. Accordingly, the heated coolant may release heat to the refrigerant passing through the refrigerant passage **39***a* of the accumulator **39** and the refrigerant passing through the refrigerant passage **38***a* of the battery chiller **38**. Thus, the refrigerant may absorb heat from the coolant in two steps through the battery chiller **38** and the accumulator **39**. As a result, the heat absorption and evaporation of the refrigerant may be significantly improved, and cabin heating performance of the HVAC subsystem **11** may be significantly improved in a condition in which the ambient temperature is relatively low.

[0077] FIG. **2** illustrates a process of a vehicle thermal management system according to an embodiment of the present disclosure in which a refrigerant absorbs heat through an accumulator, a battery chiller, and an exterior heat exchanger.

[0078] Referring to FIG. **2**, in a condition in which an ambient temperature is relatively low (i.e., in winter, in a cold area, and the like), the HVAC subsystem **11** may operate in a heating mode. The refrigerant control valve **33** may perform the first switching operation to allow first port **33***a* to be fluidly connected to the second port **33***b*. The chiller-side expansion valve **37** may perform the first switching operation to allow the first port **37***a* to be fluidly connected to the second port **37***b*. The cooling-side expansion valve **35** may be blocked. The refrigerant compressed by the compressor **31** may pass through the interior condenser **32**, and the refrigerant discharged from the interior condenser **32** may pass through the first port **33***a* and the second port **33***b* of the refrigerant control valve **33**. When the refrigerant control valve **33** performs the first switching operation, the opening degree of the second port **33***b* may be adjusted by the valve member to exceed 0% and be less than 100% so that the refrigerant discharged from the second port **33***b* of the refrigerant control valve **33** may be expanded. The refrigerant discharged from the second port **33***b* of the refrigerant control valve **33** may pass through the exterior heat exchanger **34** through the third refrigerant line **21***c*. The refrigerant passing through the exterior heat exchanger **34** may absorb heat from the ambient air. The refrigerant discharged from the exterior heat exchanger **34** may pass through the second port **37***b* and the first port **37***a* of the chiller-side expansion valve **37**. The refrigerant passing through the first port **37***a* of the chiller-side expansion valve **37** may be expanded, and the expanded refrigerant may pass through the refrigerant passage **38***a* of the battery chiller **38**. The refrigerant discharged from the refrigerant passage **38***a* of the battery chiller **38** may pass through the refrigerant passage **39***a* of the accumulator **39**, and the refrigerant discharged from the refrigerant passage **39***a* of the accumulator **39** may flow into the compressor **31**.

[0079] Referring to FIG. **2**, the coolant control valve **50** of the coolant subsystem **12** may operate to allow the second port **52** to be fluidly connected to the fourth port **54**. The distribution valve **46** may perform the second switching operation to allow the inlet port **46***a* to be fluidly connected to the second outlet port **46***c*. The coolant may be directed to the coolant heater **43** through the inlet port **46***a*, the second outlet port **46***c* of the distribution valve **46**, and the battery bypass line **23***g* by the first pump **47**. As the coolant heater **43** operates, the coolant may be heated by the coolant heater **43**, and the heated coolant may pass through the coolant passage **39***b* of the accumulator **39**. The coolant discharged from the coolant passage **39***b* of the accumulator **39** may pass through the coolant passage **38***b* of the battery chiller **38**, and the coolant discharged from the coolant passage **38***b* of the battery chiller **38** may be directed to the second port **52** of the coolant control valve **50**.

[0080] As described above, the coolant heated by the coolant heater **43** may sequentially pass through the coolant passage **39***b* of the accumulator **39** and the coolant passage **38***b* of the battery chiller **38**. Accordingly, the heated coolant may release heat to the refrigerant passing through the refrigerant passage **39***a* of the accumulator **39** and the refrigerant passing through the refrigerant passage **38***a* of the battery chiller **38**. In addition, the refrigerant may absorb heat from the ambient air in the exterior heat exchanger **34**. Thus, the refrigerant may absorb heat from the coolant in two steps through the battery chiller **38** and the accumulator **39** and the refrigerant may absorb heat from the ambient air through the exterior heat exchanger **34**. As a result, heat absorption and evaporation of the refrigerant may be significantly improved, and cabin heating performance of the HVAC subsystem **11** may be significantly improved in a condition in which the ambient temperature is relatively low.

[0081] As set forth above, the vehicle thermal management system according to the embodiments of the present disclosure may be designed to improve heating performance by transferring heat of the coolant heater to the accumulator in a condition in which the ambient temperature is relatively low (i.e., in winter, in a cold area, and the like).

[0082] According to embodiments of the present disclosure, the coolant circulation path may be fluidly connected to the coolant heater, the coolant passage of the accumulator, and the coolant passage of the battery chiller so that the coolant heated by the coolant heater may sequentially pass through the coolant passage of the accumulator and the coolant passage of the battery chiller. Accordingly, the heated coolant may release heat to the refrigerant passing through the refrigerant passage of the accumulator and the refrigerant passing through the refrigerant passage of the battery chiller. Thus, the refrigerant may absorb heat from the coolant in two steps through the battery chiller and the accumulator so that the heat absorption and evaporation of the refrigerant may be significantly improved.

[0083] Although the present disclosure has been described with reference to the embodiments and the accompanying drawings, the present disclosure is not limited thereto, but may be variously modified and altered by those having ordinary skill in the art to which the present disclosure pertains without departing from the spirit and scope of the present disclosure claimed in the following claims.

What is claimed is:

1. A vehicle thermal management system, comprising:

a heating, ventilation, and air conditioning (HVAC) subsystem including a compressor, an accumulator disposed on an upstream side of the compressor, and a refrigerant circulation path fluidly connected to the accumulator; and

a coolant subsystem including a coolant heater and a coolant circulation path fluidly connected to the coolant heater,

wherein the accumulator includes a refrigerant passage fluidly connected to the refrigerant circulation path and includes a coolant passage fluidly connected to the coolant circulation path.

2. The vehicle thermal management system according to claim **1**, wherein the coolant heater is located on an upstream side of the coolant passage of the accumulator.

3. The vehicle thermal management system according to claim **1**, further comprising a battery chiller including a refrigerant passage fluidly connected to the refrigerant circulation path and including a coolant passage fluidly connected to the coolant circulation path.

4. The vehicle thermal management system according to claim **3**, wherein the refrigerant passage of the battery chiller is located on an upstream side of the refrigerant passage of the accumulator, and wherein the coolant passage of the battery chiller is located on a downstream side of the coolant passage of the accumulator.

5. The vehicle thermal management system according to claim **3**, wherein:

the HVAC subsystem further includes an interior condenser disposed on a downstream side of the compressor, a refrigerant control valve disposed on a downstream side of the interior condenser, an exterior heat exchanger disposed on a downstream side of the refrigerant control valve, and a chiller-side expansion valve disposed on the upstream side of the refrigerant passage of the battery chiller, and

the chiller-side expansion valve is configured to allow a refrigerant to flow from any one of the exterior heat exchanger and the interior condenser to the refrigerant passage of the battery chiller.

6. The vehicle thermal management system according to claim **5**, wherein the refrigerant circulation path includes a heating-side bypass line extending from an upstream point of the refrigerant control valve to an upstream point of the refrigerant passage of the battery chiller.

7. The vehicle thermal management system according to claim **6**, wherein the chiller-side expansion valve includes a first port fluidly communicating with the refrigerant passage of the battery chiller, a second port fluidly communicating with the exterior heat exchanger, and a third port fluidly communicating with the heating-side bypass line.

8. The vehicle thermal management system according to claim **7**, wherein the chiller-side expansion valve is configured to allow the first port to be fluidly connected to at least one of the second port and the third port by an actuator.

9. The vehicle thermal management system according to claim **3**, wherein the coolant subsystem further includes a battery and a coolant control valve fluidly connected to the coolant circulation path, and wherein the coolant control valve is configured to control a flow of a coolant passing through the battery, the coolant heater, the coolant passage of the accumulator, and the coolant passage of the battery chiller.

10. The vehicle thermal management system according to claim **9**, wherein the coolant heater is located on a downstream side of the battery.

11. The vehicle thermal management system according to claim **10**, wherein the coolant subsystem further includes a battery bypass line extending from an upstream point of the battery to a downstream point of the battery.

12. The vehicle thermal management system according to claim **11**, wherein the coolant subsystem further includes a distribution valve configured to allow the coolant discharged from the coolant control valve to be directed to at least one of the battery and the battery bypass line.

13. The vehicle thermal management system according to claim **12**, wherein the distribution valve includes an inlet port fluidly communicating with the coolant control valve, a first outlet port fluidly communicating with the battery, and a second outlet port fluidly communicating with the battery bypass line.

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